MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY TTL, FLIP-FLOPS, CASCADABLE, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, bipolar, low-power Schottky TTL, flip-flops, bistable logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided
and are reflected in the complete part number.

1.2 Part number. The complete part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. Device type shall be as shown in the following:

Device type	Circuit
01	Octal D flip-flop with clear, cascadable
02	Octal D flip-flop with transparent latch and 3-state outputs non-cascadable
03	Octal D flip-flop with three-state outputs, cascadable
04	Octal D flip-flop with enable, cascadable

1.2.2 Device class. Device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
R	D-8 (20-lead, 1/4" x 1 1/16"), dual-in-line package
S	F-9 (20-lead, 1/4" x 1/2"), flat package
2	C-2 (20 terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to 7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to 5.5 V dc
Storage temperature range	-65°C to 150°C

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Rome Air Development Center, RBE-2, Griffiss AFB, NY 13441 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Power dissipation, (P_D) 1/	
Device type	
01	149 mW
02, 03	231 mW
04	176 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC})	
Cases R, S	(See MIL-M-38510, appendix C)
Case 2	60°C/W 2/
Junction temperature (T_J) 3/	+175°C

1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH})	2.0 V dc
Maximum low level input voltage (V_{IL})	0.7 V dc
Case operating temperature range (T_C)	-55°C to +125°C
Normalized fanout (each output) 4/	10 maximum at low logic level 20 maximum at high logic level
Input setup time, $t_{(setup)}$	
Device type	
01, 03	20 ns minimum
02	5 ns minimum
04	
data	20 ns minimum
enable active	25 ns minimum
enable inactive	10 ns minimum
Input hold time, $t_{(hold)}$	
Device type	
01, 03, 04	5 ns minimum
02	20 ns minimum
Input pulse width, t_p	
Device type	
01, 04 (clock)	20 ns minimum
01 (clear)	20 ns minimum
02 (enable)	15 ns minimum
03 (clock)	15 ns minimum

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).
2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein.
3/ Maximum junction temperature shall not be exceeded except for short duration burn-in screening conditions per method 5004 of MIL-STD-883.
4/ Device will fanout in both high and low levels to the specified number of data inputs on the same device as that being tested.

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(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a pre-requisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are specified in table I and apply over the full recommended case operating temperature range.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 10 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007 of MIL-STD-883, as applicable, except as modified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$		Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{IH} = 2.0 V, V_{CC} = 4.5 V	I_{OH} = -400 µA	01,04	2.5	---	V
			I_{OH} = -1 mA	02,03	2.4	---	V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IL} = 0.7 V	I_{OL} = 4 mA	01,04	---	0.4	V
			I_{OL} = 12 mA	02,03			
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{OL} = -18 mA, T_C = 25°C		All	---	-1.5	V
Low-level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	clear	01	-30	-360	µA
			clock	01,04	-30	-400	
				03	0	-400	
			data	01,04	-100	-400	
				02	0	-400	
				03	0	-360	
			D/C	02,03	0	-420	
			enable	02	---	-400	
				04	-30	-400	
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V		All	---	20	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		All	---	100	µA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{IN} = 0 V 1/		02,03	-15	-225	mA
				01,04	-15	-100	
Output current, high level, outputs off	I_{OZH}	V_{CC} = 5.5 V, V_O = 2.7 V		02,03	---	20	µA
Output current, low level, outputs off	I_{OZL}	V_{CC} = 5.5 V, V_O = 0.4 V		02,03		-20	µA

1/ Not more than one output should be shorted at a time.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$		Device type	Limits Min	Limits Max	Unit
Supply current	I_{CC}	V_{CC} = 5.5 V	V_{IN} = 5.0 V	01	---	27	mA
				02	---	40	mA
				03		42	
			V_{IN} = GND	04	---	32	mA
Maximum clock frequency	f_{MAX}	V_{CC} = 4.5 V C_L = 50 pF ±10%	R_L = 2 kΩ ±5%	01,04	25	---	MHz
			R_L = 110Ω ±5%	03	26	---	MHz
To low level clr to Q	t_{PHL1}	C_L = 50 pF ±10% V_{CC} = 5.0 V R_L = 2 kΩ ±5%		01	5	42	ns
To high level clk to Q	t_{PLH2}			01	5	42	ns
To low level clk to Q	t_{PHL2}			01	5	42	ns
To high level D to Q	t_{PLH1}	V_{CC} = 5.0 V C_L = 50 pF ±10% R_L = 110Ω ±5%		02	3	24	ns
To low level D to to Q	t_{PHL1}			02	3	24	ns
To high level G to Q	t_{PLH2}			02	3	39	ns
To low level G to to Q	t_{PHL2}			02	3	39	ns
To high level O/C to Q	t_{PZH}			02	3	37	ns
To low level O/C to Q	t_{PZL}			02	3	47	ns
To Z from high O/C to Q	t_{PHZ}			02	3	35	ns
To Z from low O/C to Q	t_{PLZ}			02	3	39	ns
To high level clk to Q	t_{PLH2}			03	5	37	ns
To low level clk to Q	t_{PHL2}			03	5	37	ns
To high level O/C to Q	t_{PZH}			03	3	37	ns
To low level O/C to Q	t_{PZL}			03	3	37	ns

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits		Unit
				Min	Max	
To Z from high O/C to Q	t_{PHZ}	C_L = 50 pF ±10% V_{CC} = 5.0 V R_L = 110Ω ±5%	03	3	37	ns
To Z from low O/C to Q	t_{PLZ}		03	3	39	ns
To high level clk to Q	t_{PLH2}	C_L = 50 pF ±10% R_L = 2 kΩ ±5% V_{CC} = 5.0 V	04	5	42	ns
To low level clk to Q	t_{PHL2}		04	5	42	ns

TABLE II. Electrical test requirements.

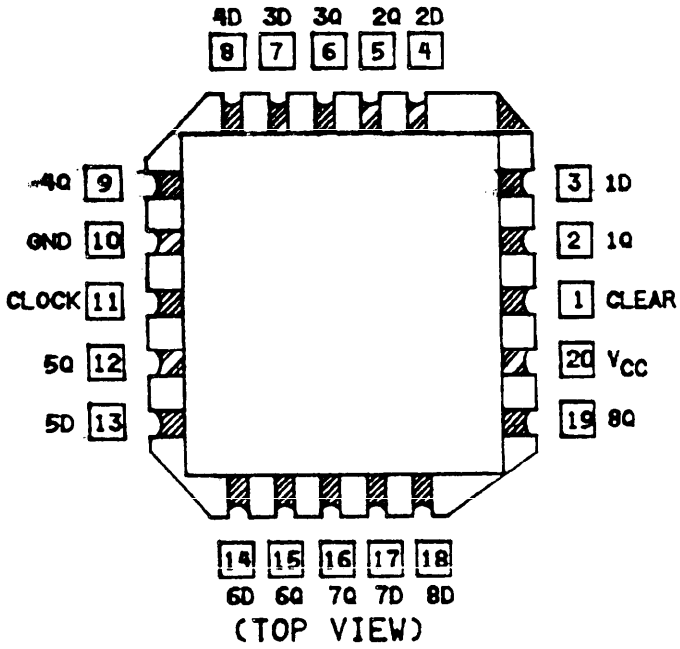
MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*, 2, 3, 7, 9, 10, 11	1*,2,3,7,9
Group A test requirements (method 5005)	1, 2, 3, 7, 8, 9, 10, 11	1,2,3,7,8,9
Group B test requirements (method 5005) subgroup 5	1, 2, 3, 7, 8, 9, 10, 11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1, 2, 3
Additional electrical subgroups for group C periodic inspections	N/A	10, 11
Group D end-point electrical parameters (method 5005)	1, 2, 3	1, 2, 3

*PDA applies to subgroup 1 (see 4.3c).

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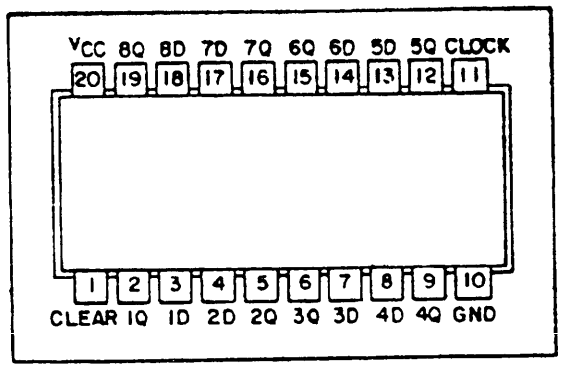
Device type 01

Case 2



Device type 01

Cases R and S



Device type 01

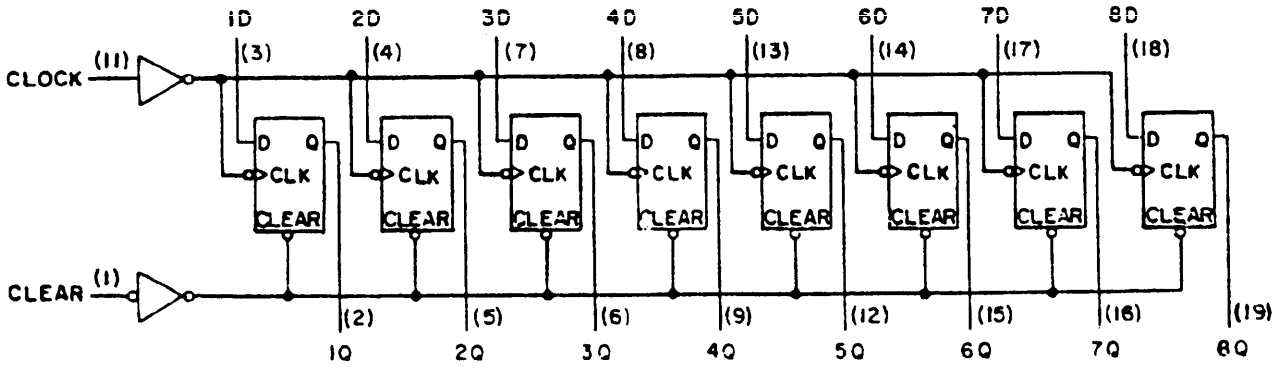
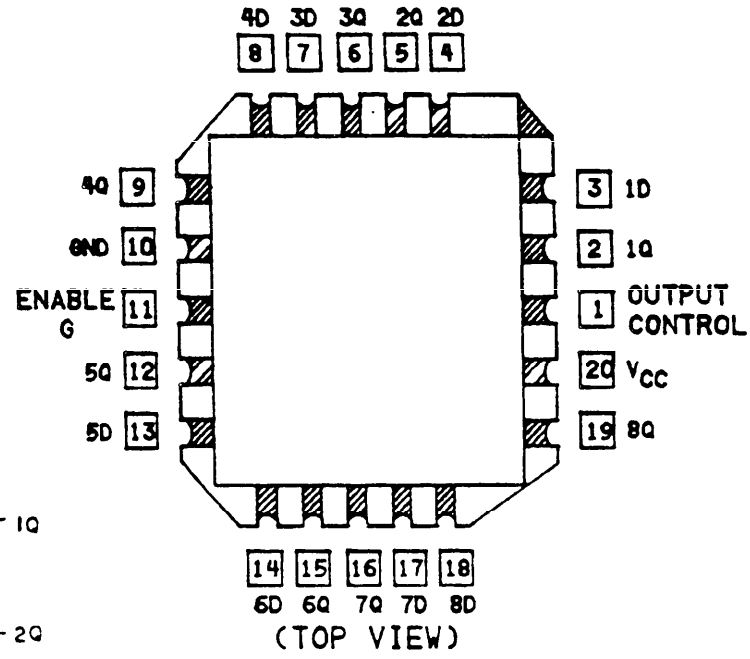


FIGURE 1. Logic diagrams and terminal connections.

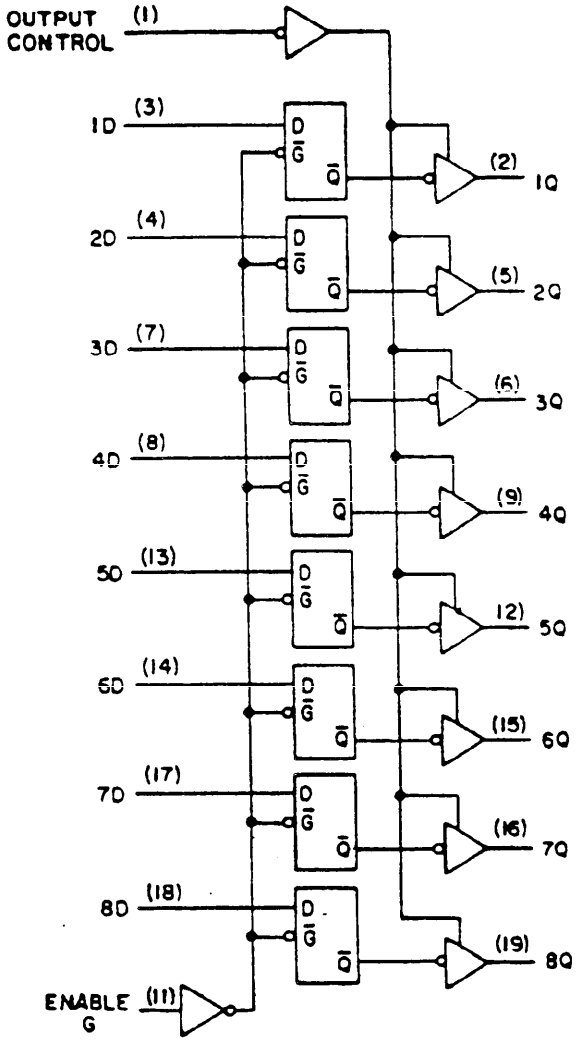
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Device type 02

Case 2



Device type 02



Device type 02

Cases R and S

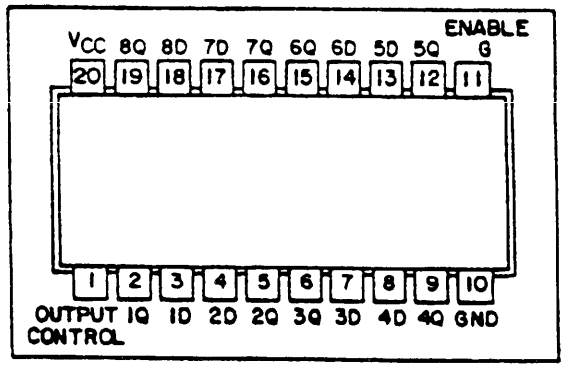
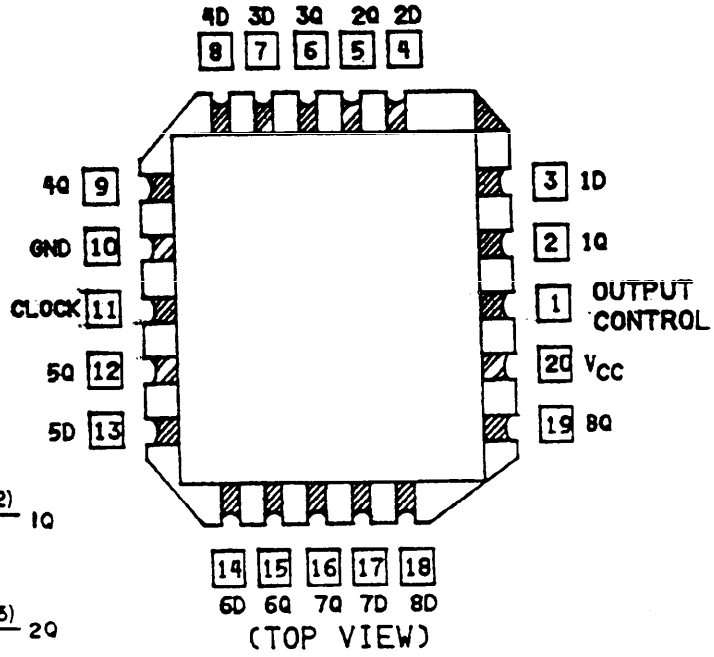


FIGURE 1. Logic diagrams and terminal connections - Continued.

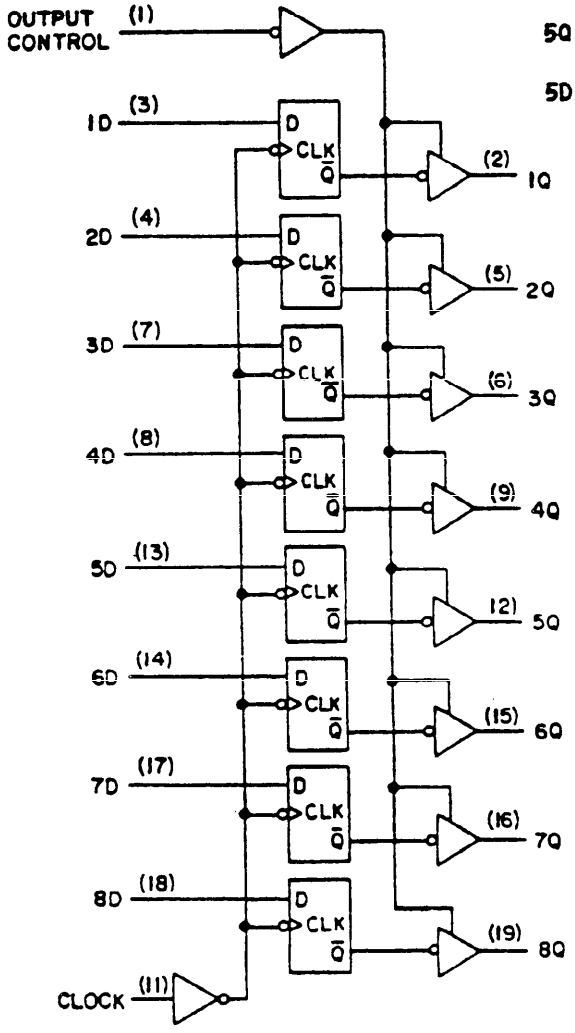
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Device type 03

Case 2



Device type 03



Device type 03

Cases R and S

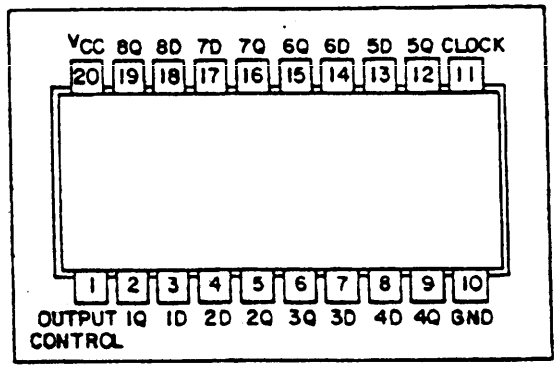
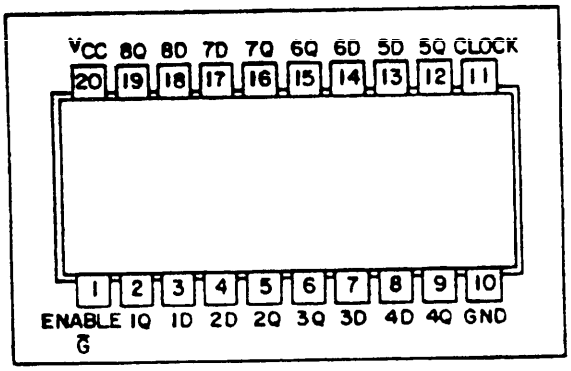


FIGURE 1. Logic diagrams and terminal connections - Continued.

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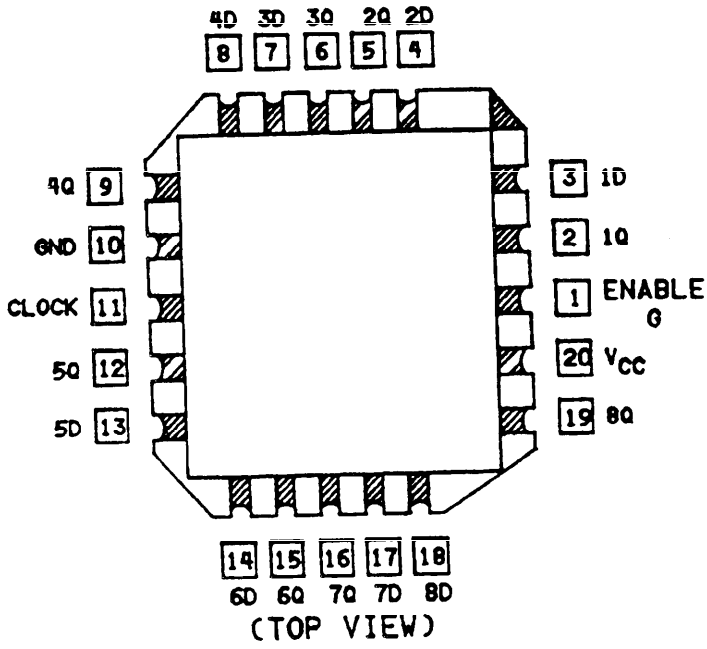
Device type 04

Cases R and S



Device type 04

Case 2



Device type 04

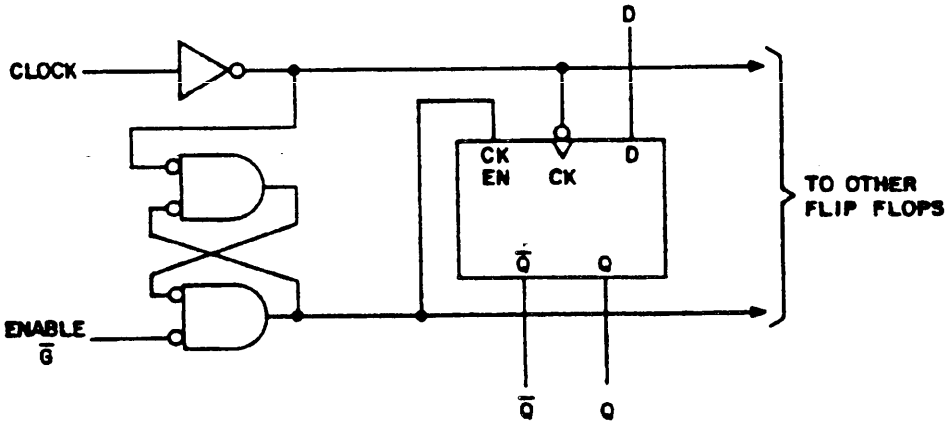


FIGURE 1. Logic diagrams and terminal connections - Continued.

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Device type 01

Inputs			Output
Clear	Clock	D	Q
L	X	X	L
H	↑	H	H
H	↑	L	L
H	L	X	Q_0

H = high level (steady state)
L = low level (steady state)
X = irrelevant
↑ = transition from low to high level
Q_0 = the level of Q before the indicated input conditions were established

Device type 02

Output control	Enable G	D	Output
L	H	H	H
L	H	L	L
L	L	X	Q_0
H	X	X	Z

H = high level (steady state)
L = low level (steady state)
X = irrelevant
Q_0 = the level of Q before the indicated steady state input conditions were established.
Z = high impedance state

FIGURE 2. Truth tables.

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Device type 03

Output control	Clock	D	Output
L		H	H
L		L	L
L	L	X	Q_0
H	X	X	Z

H = high level (steady state)
L = low level (steady state)
X = irrelevant
  = transition from low to high level
Q_0 = the level of Q before the indicated steady state input conditions were established
Z = high impedance state

Device type 04

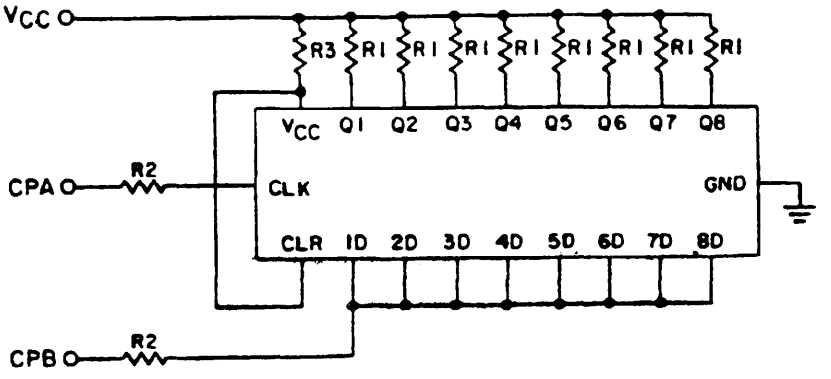
Inputs			Output
G	Clock	Data	Q
H	X	X	Q_0
L		H	H
L		L	L
X	L	X	Q_0

H = high level (steady state)
L = low level (steady state)
X = irrelevant
  = transition from low to high level
Q_0 = the level of Q before the indicated steady state input conditions were established

FIGURE 2. Truth tables - Continued.

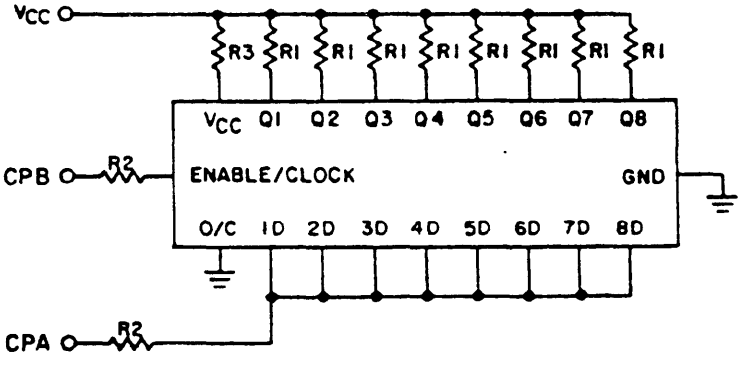
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Device type 01



NOTES:
1. CPA = 100 kHz ±50 percent square wave; duty cycle = 50 ±15 percent; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
   CPB = Same as CPA, synchronized with CPA, except 50 kHz ±50 percent square wave.
2. R1 = 1 kΩ ±5 percent; R2 = 27Ω maximum; R3 = 10Ω maximum.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

Device types 02 and 03

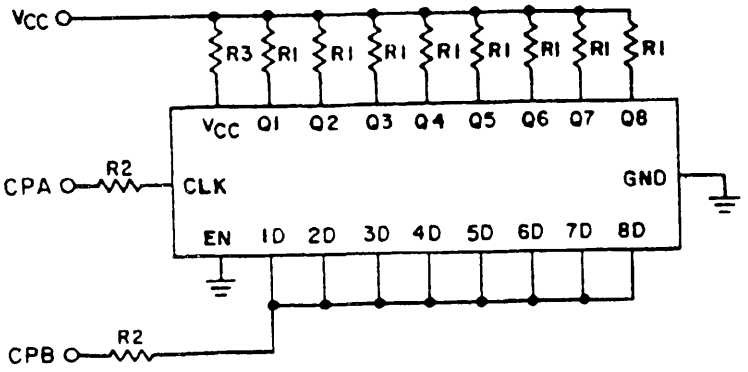


NOTES:
1. CPA = 100 kHz ±50 percent square wave; duty cycle = 50 ±15 percent; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
   CPB = Same as CPA, synchronized with CPA, except 50 kHz ±50 percent square wave.
2. R1 = 470Ω ±5 percent; R2 = 27Ω maximum; R3 = 10Ω maximum.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

FIGURE 3. Dynamic burn-in and steady-state life test circuit.

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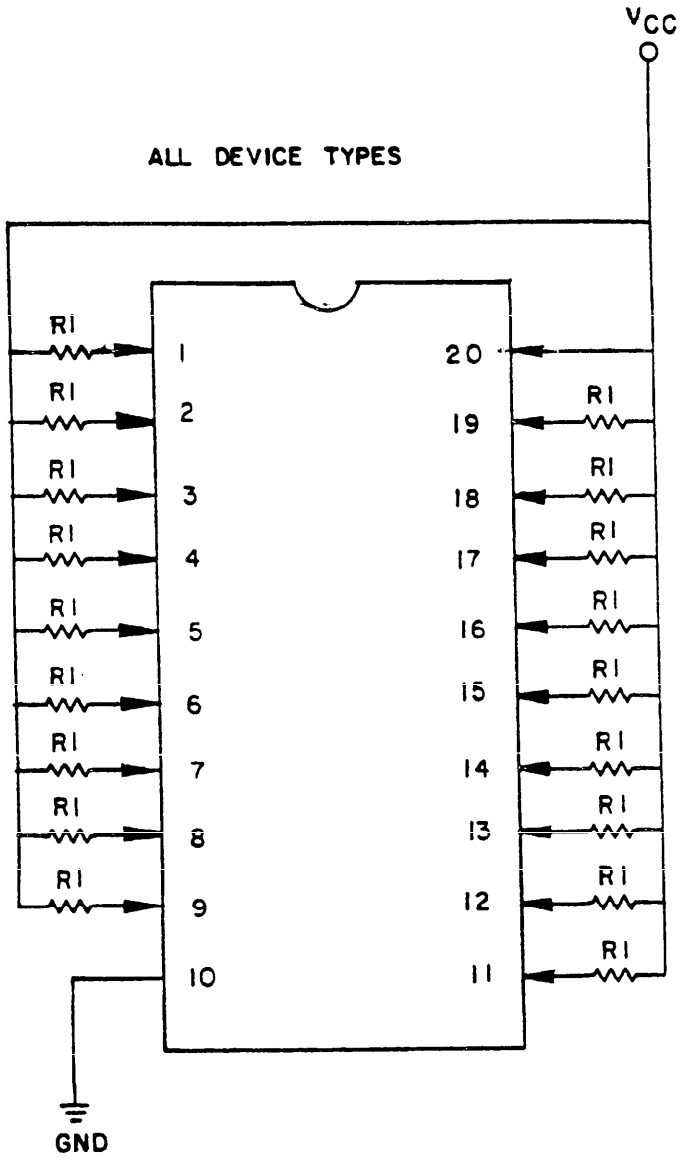
Device type 04



NOTES:
1. CPA = 100 kHz ±50 percent square wave; duty cycle = 50 ±15 percent; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
   CPB = Same as CPA, synchronized with CPA, except 50 kHz ±50 percent square wave.
2. R1 = 1 kΩ ±5 percent; R2 = 27Ω maximum; R3 = 10Ω maximum.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

FIGURE 3. Dynamic burn-in and steady-state life test circuit - Continued.

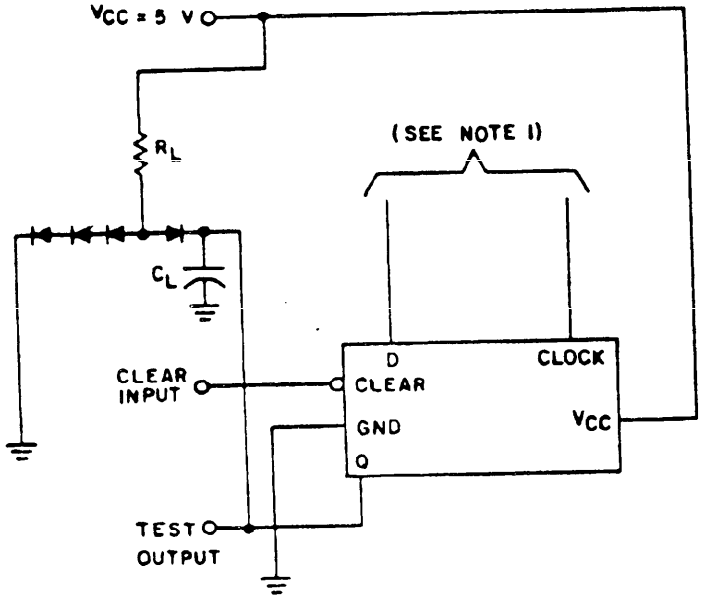
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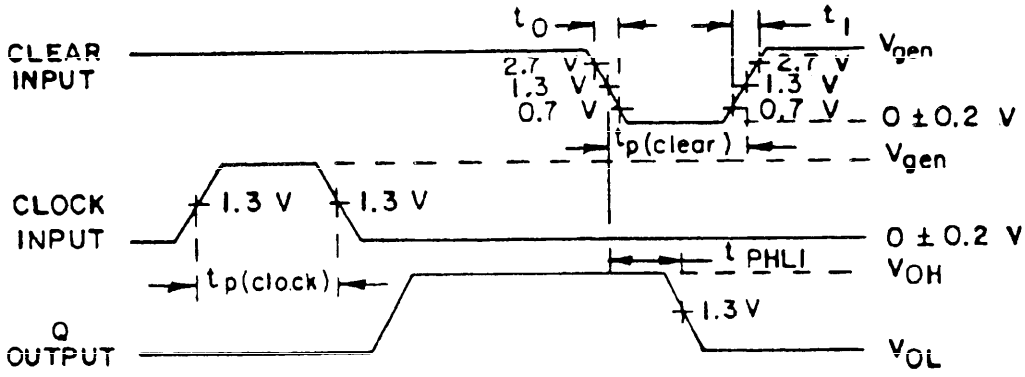
NOTES:
1. High temperature (+175°C) condition F burn-in circuit only.
2. RI = 2 kΩ.
3. V_{CC} shall be chosen such that 5.0 V minimum is present at V_{CC} device terminal.

FIGURE 3. Dynamic burn-in steady-state life test circuit - Continued.

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TEST CIRCUIT

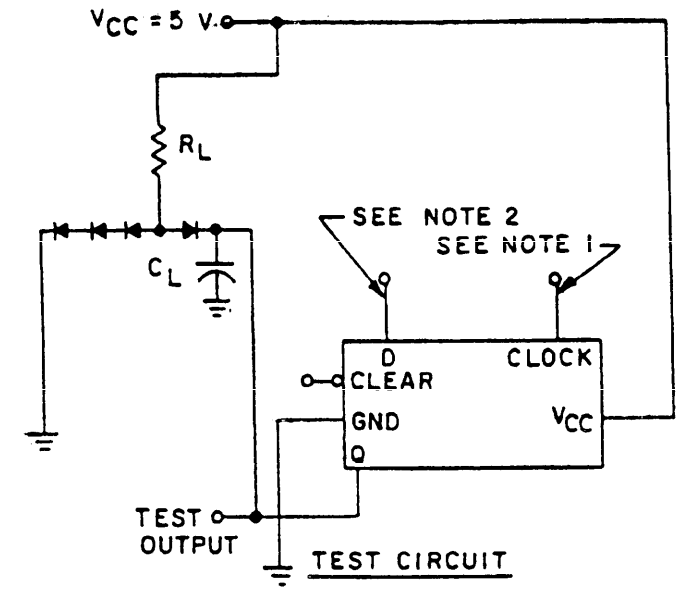


VOLTAGE WAVEFORMS

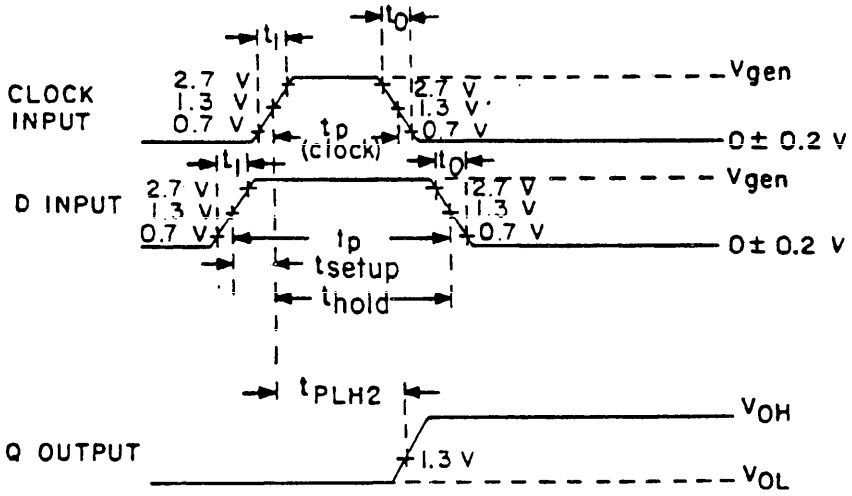
NOTES:
1. Clear input dominates regardless of the state of clock or D inputs.
2. All diodes are 1N3064, or equivalent.
3. Clear input pulse characteristics: V_{gen} = 3 ±0.2 V, $t_0 \leq 6$ ns, $t_1 \leq 15$ ns, t_p (clear) = 20 ns minimum and PRR ≤ 1 MHz.
4. C_L = 50 pF ±10 percent (including jig and probe capacitance).
5. R_L = 2 kΩ ±5 percent.
6. Clock input pulse characteristics: t_p (clock) ≥ 20 ns; V_{gen} = 3 ±0.2 V; PRR ≤ 1 MHz.

FIGURE 4. Asynchronous switching test circuit for device type 01.

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TEST CIRCUIT

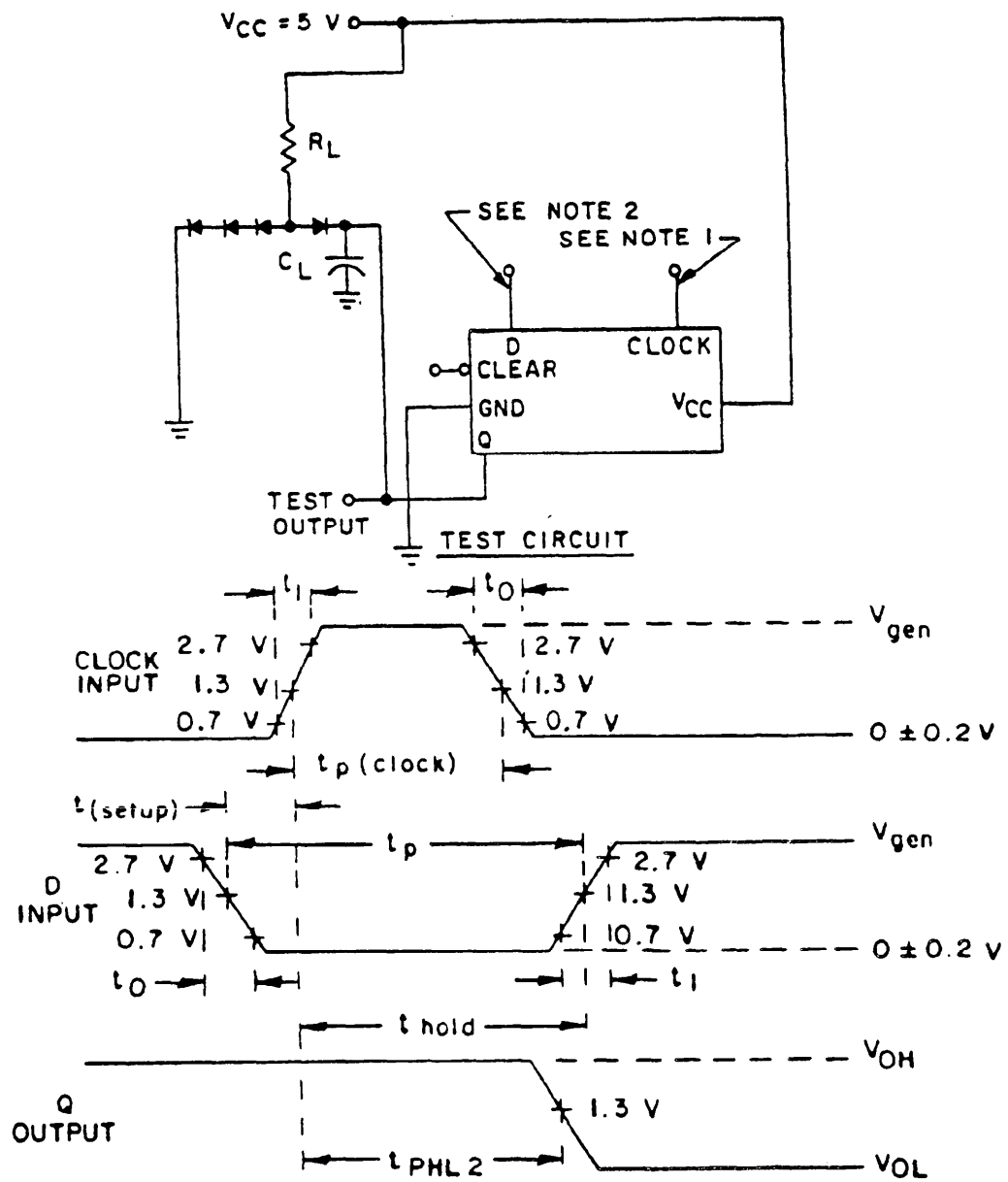


VOLTAGE WAVEFORMS

NOTES:
1. Clock input pulse has the following characteristics: V_{gen} = 3 ±0.2 V, $t_1 \leq 15$ ns, $t_0 \leq 6$ ns, t_p (clock) = 30 ns, and PRR ≤ 1 MHz. When testing f_{MAX}, PRR = see table III, and $t_0 = t_1 \leq 6$ ns.
2. D input has the following characteristics: V_{gen} = 3 ±0.2 V, $t_1 \leq 15$ ns, $t_0 \leq 6$ ns, t_{setup} = 20 ns, t_{hold} = 5 ns, t_p = 25 ns, and PRR is 50 percent of the clock PRR. For f_{MAX}, $t_0 = t_1 \leq 6$ ns.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 pF ±10 percent (including jig and probe capacitance).
5. R_L = 2 kΩ ±5 percent.

FIGURE 4. Synchronous switching test circuit (high-level data) for device types 01 and 04 - Continued.

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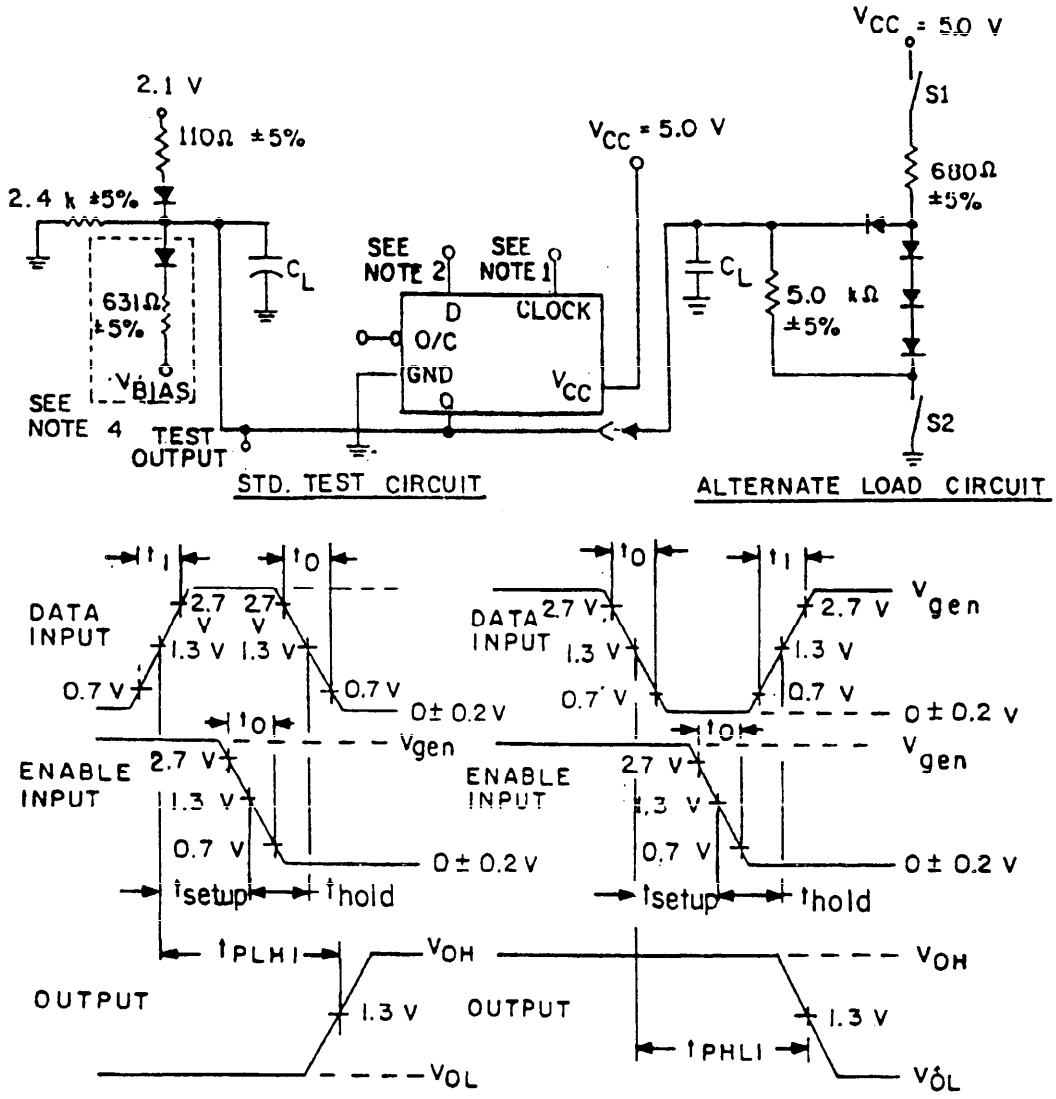
VOLTAGE WAVEFORMS

NOTES:

1. Clock input pulse has the following characteristics: V_{gen} = 3 ±0.2 V, $t_1 \leq 15$ ns, $t_0 \leq 6$ ns, t_p (clock) = 30 ns, and PRR ≤ 1 MHz.
2. D input has the following characteristics: V_{gen} = 3 ±0.2 V, $t_1 \leq 15$ ns, $t_0 \leq 6$ ns, t_{setup} = 20 ns, t_{hold} = 5 ns, t_p = 25 ns, and PRR is 50 percent of the clock PRR.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 pF ±10 percent (including jig and probe capacitance).
5. R_L = 2 kΩ ±5 percent.

FIGURE 4. Synchronous switching test circuit (low-level data) for device types 01 and 04 - Continued.

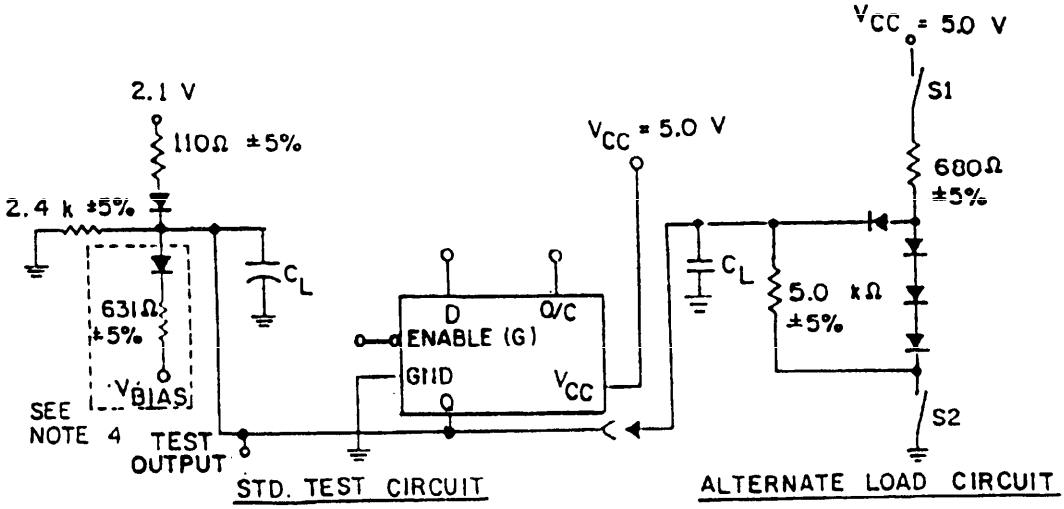
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:12Z
Check the source to verify that this is the current version before use.

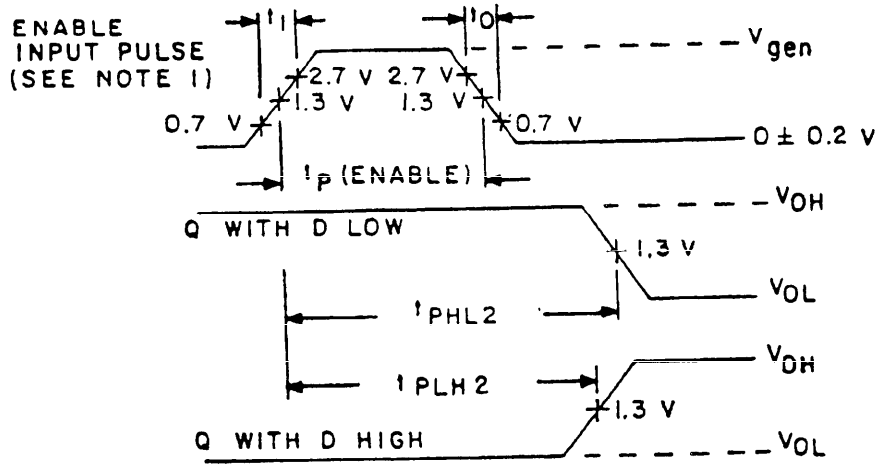


NOTES:
1. Enable pulse has the following characteristics: V_{gen} = 3 ±0.2 V, $t_0 \leq 6$ ns.
2. D input pulse has the following characteristics: V_{gen} = 3 ±0.2 V, $t_1 \leq 15$ ns, $t_0 \leq 6$ ns, t_{setup} = 5 ns, t_{hold} = 20 ns, t_p = 25 ns.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 pF ±10 percent (including jig and probe capacitance).
5. The diode and resistor shown within the dotted area are optional. When the diode and resistor are used, V_{BIAS} shall be 5.5 V for all tests except for t_{PHZ}; for t_{PHZ} tests, V_{BIAS} shall be -0.6 V.
6. Switches S1 and S2 are closed for t_{PLH1} and t_{PHL1}, when using the alternate load circuit.

FIGURE 4. Switching test circuit (t_{PLH1}, t_{PHL1}) for device type 02 - Continued.

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VOLTAGE WAVEFORMS

NOTES:
1. Enable input characteristics for t_{PLH2} and t_{PHL2}, V_{gen} = 3 ±0.2 V, $t_0 \leq 6$ ns, $t_1 \leq 15$ ns, t_p (enable) = 15 ns, and PRR ≤ 1 MHz.
2. All diodes are 1N3064, or equivalent.
3. C_L = 50 pF ±10 percent (including jig and probe capacitance).
4. The diode and resistor shown within the dotted area are optional. When the diode and resistor are used, V_{BIAS} shall be 5.5 V for all tests except for t_{PHZ}; for t_{PHZ} tests, V_{BIAS} shall be -0.6 V.
5. Switches S1 and S2 are closed for t_{PLH2} and t_{PHL2} when using the alternate load circuit

FIGURE 4. Switching test circuit (t_{PLH2}, t_{PHL2}) for device type 02 - Continued.

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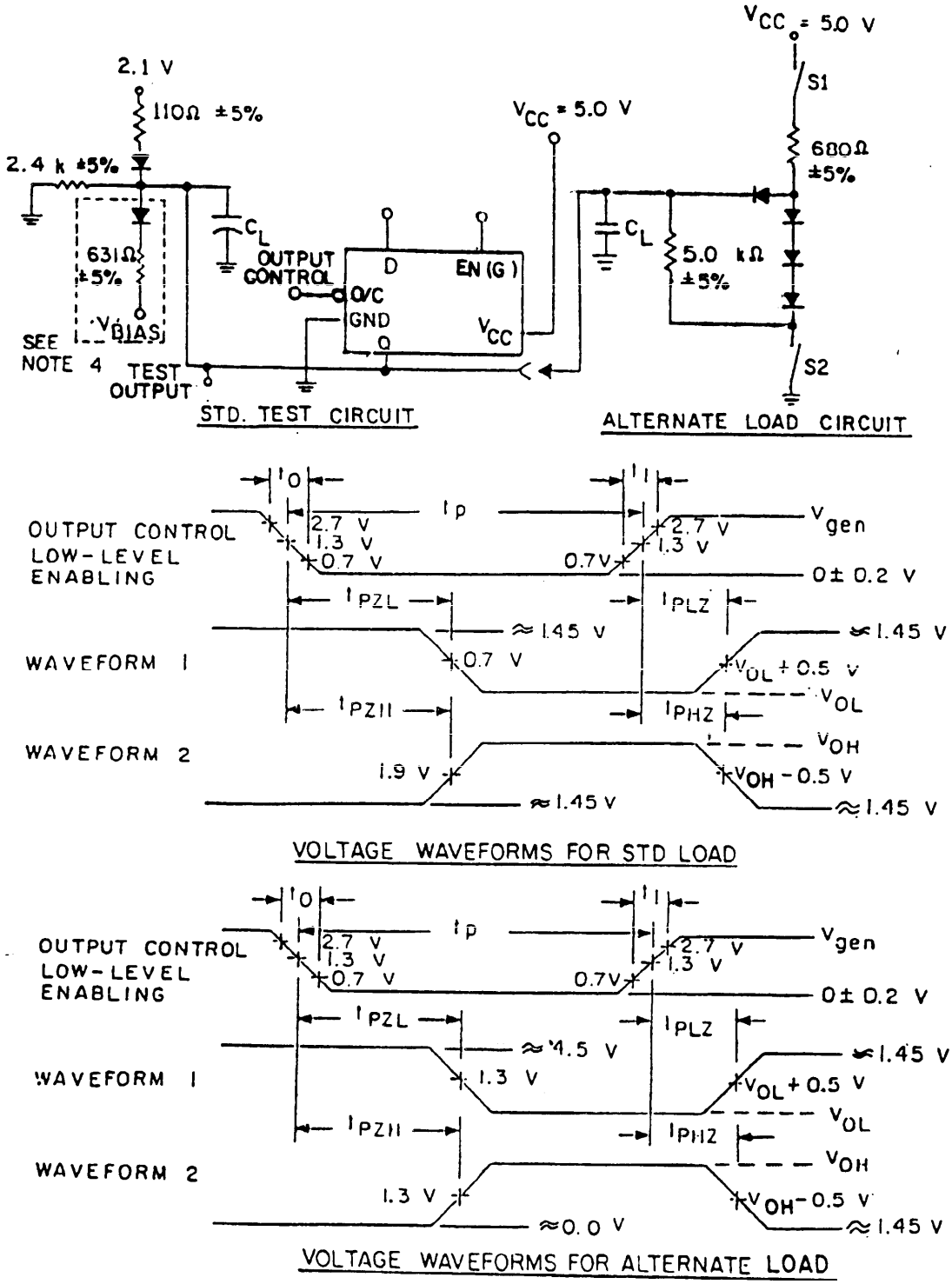


FIGURE 4. Three-state switching test circuits and waveforms for device type 02 - Continued.

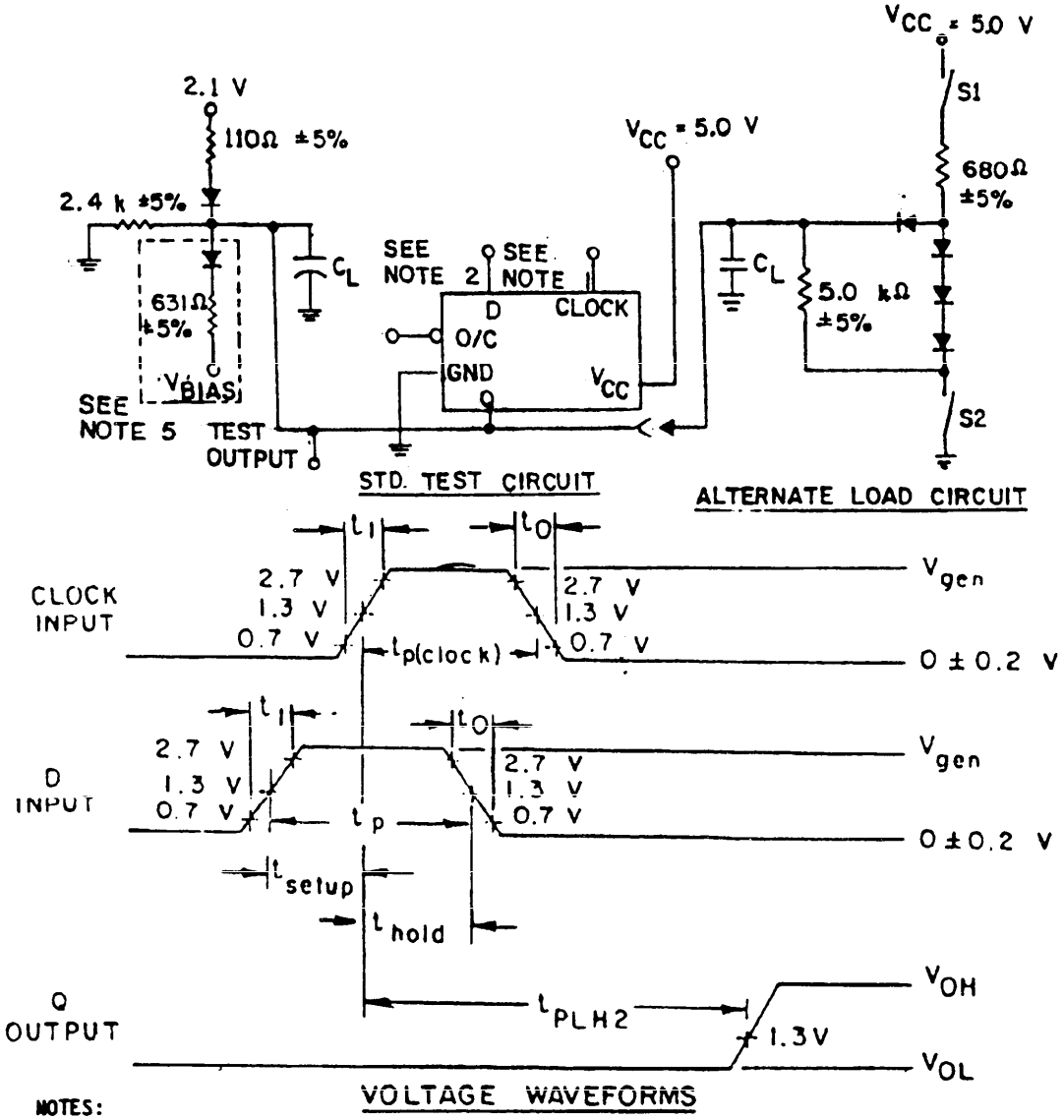
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:12Z
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NOTES:
1. All diodes are 1N3064, or equivalent.
2. Output control input pulse characteristics: V_{gen} = 3 ±0.2 V, $t_0 \leq 6$ ns, $t_1 \leq 15$ ns, t_p (input) $\geq$ 200 ns, and PRR $\leq$ 1 MHz.
3. For the standard test circuit, C_L = 50 pF ±10 percent. For the alternate load circuit, C_L = 50 pF ±10 percent for t_{PZL} and t_{PZH} and C_L = 15 pF minimum for t_{PLZ} and t_{PHZ}. All capacitive load values include jig and probe capacitance.
4. The diode and resistor shown within the dotted area are optional. When the diode and resistor are used, V_{BIAS} shall be 5.5 V for all tests except for t_{PHZ}; for t_{PHZ} tests, V_{BIAS} shall be -0.6 V.
5. When using the alternate load circuit, the appropriate switch positions for the three-state tests are listed in the table below.

Test	S1	S2
t_{PZL}	Closed	Open
t_{PLZ}	Closed	Closed
t_{PZH}	Open	Closed
t_{PHZ}	Closed	Closed

FIGURE 4. Three-state switching test circuit and waveforms for device type 02 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:12Z
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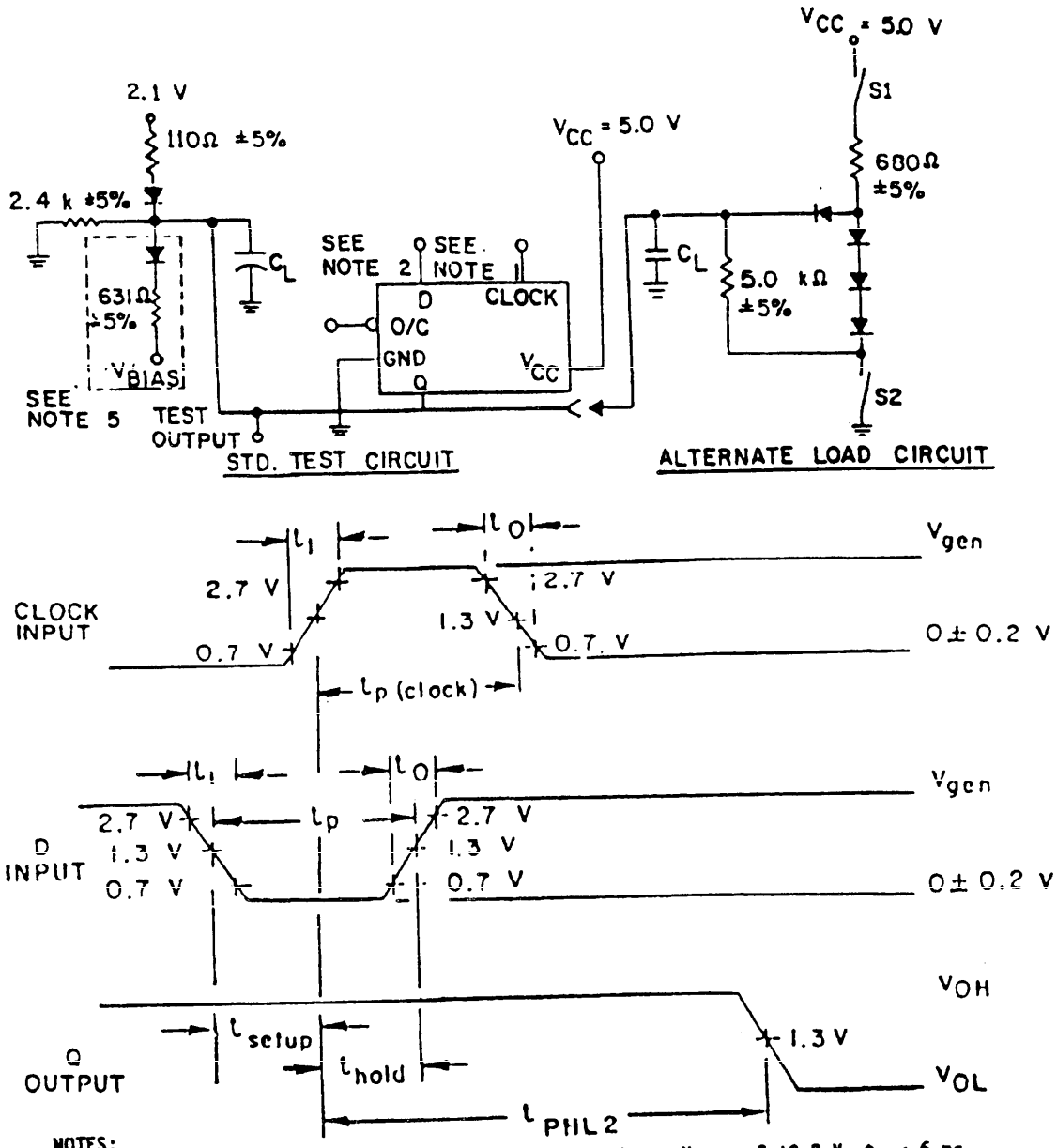


NOTES:

1. Clock input pulse has the following characteristics: V_{gen} = 3 ±0.2 V, $t_0 \leq 6$ ns, $t_1 \leq 15$ ns, t_p (clock) = 15 ns, and PRR ≤ 1 MHz. When testing f_{MAX}, PRR = see table III, and $t_0 = t_1 \leq 6$ ns.
2. D input pulse has the following characteristics: V_{gen} = 3 ±0.2 V, $t_0 \leq 6$ ns, $t_1 \leq 15$ ns, t_{setup} = 20 ns, t_{hold} = 5 ns, t_p = 25 ns, and PRR = 50 percent of the clock PRR. For f_{MAX}, $t_0 = t_1 \leq 6$ ns.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 pF ±10 percent (includes jig and probe capacitance).
5. The diode and resistor shown within the dotted area are optional. When the diode and resistor are used, V_{BIAS} shall be 5.5 V for all tests except for t_{PHZ}, for t_{PHZ} tests, V_{BIAS} shall be -0.6 V.
6. Switches S1 and S2 are closed for t_{PHL2}, when using the alternate load circuit.

FIGURE 4. Synchronous switching test circuit (high-level data) for device type 03 - Continued.

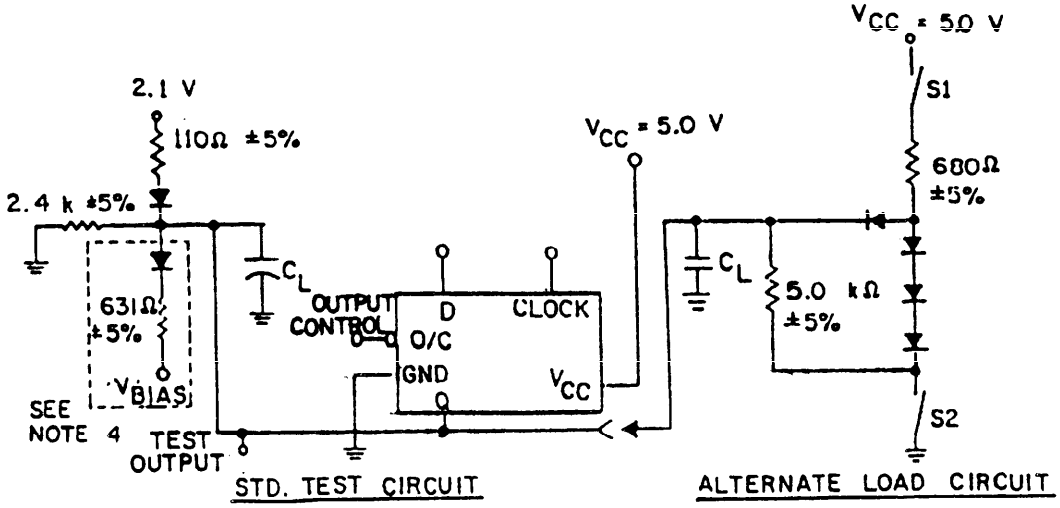
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:12Z
Check the source to verify that this is the current version before use.

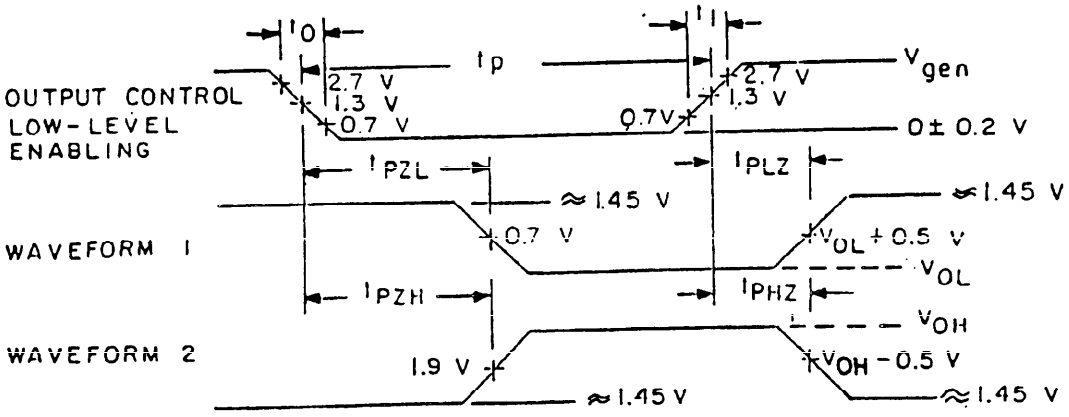


NOTES:
1. Clock input pulse has the following characteristics: V_{gen} = 3 ±0.2 V, $t_0 \leq 6$ ns, $t_1 \leq 15$ ns, t_p (clock) = 15 ns, and PRR ≤ 1 MHz.
2. D input pulse has the following characteristics: V_{gen} = 3 ±0.2 V, $t_0 \leq 6$ ns, $t_1 \leq 15$ ns, t_{setup} = 20 ns, t_{hold} = 5 ns, t_p = 25 ns, and PRR = 50 percent of the clock PRR.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 pF ±10 percent (includes jig and probe capacitance).
5. The diode and resistor shown within the dotted area are optional. When the diode and resistor are used, V_{BIAS} shall be 5.5 V for all tests except for t_{PHZ}, for t_{PHZ} tests, V_{BIAS} shall be -0.6 V.
6. Switches S1 and S2 are closed for t_{PHL2}, when using the alternate load circuit.

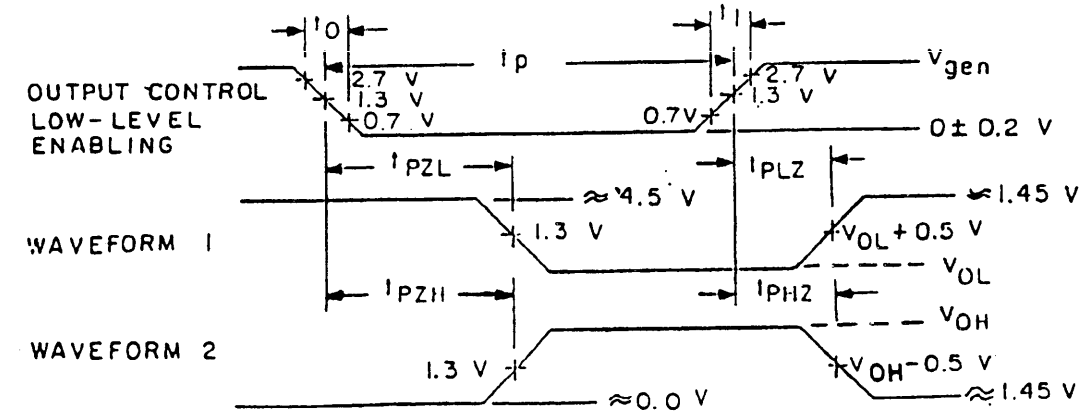
FIGURE 4. Synchronous switching test circuit (low-level data) for device type 03 - Continued.

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VOLTAGE WAVEFORMS FOR STD LOAD



VOLTAGE WAVEFORMS FOR ALTERNATE LOAD

FIGURE 4. Three-state switching test circuits and waveforms for device type 03 - Continued.

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NOTES:
1. O/C input pulse characteristics: V_{gen} = 3 ±0.2 V, $t_0 \leq 6$ ns, $t_1 \leq 15$ ns, t_p (input) $\geq$ 200 ns, and PRR $\leq$ 1 MHz.
2. All diodes are 1N3064, or equivalent.
3. For the standard test circuit, C_L = 50 pF ±10 percent. For the alternate load circuit, C_L = 50 pF ±10 percent for t_{PZL} and t_{PZH} and C_L = 15 pF minimum for t_{PLZ} and t_{PHZ}. All capacitive load values include jig and probe capacitance.
4. The diode and resistor shown within the dotted area are optional. When the diode and resistor are used, V_{BIAS} shall be 5.5 V for all tests except for t_{PHZ}; for t_{PHZ} tests, V_{BIAS} shall be -0.6 V.
5. When using the alternate load circuit, the appropriate switch positions for the three-state tests are listed in the table below.

Test	S1	S2
t_{PZL}	Closed	Open
t_{PLZ}	Closed	Closed
t_{PZH}	Open	Closed
t_{PHZ}	Closed	Closed

FIGURE 4. Three-state switching test circuits and waveforms for device type 03 - Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥ 2.0 V, or low ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases IR, S, 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				CLR	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	CLK	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1 2 3 4 5 6 7 8	2.0 V	-.4 mA	2.0 V	2.0 V	-.4 mA	-.4 mA	2.0 V	2.0 V	-.4 mA	GND	1/	-.4 mA	2.0 V	2.0 V	-.4 mA	-.4 mA	2.0 V	2.0 V	-.4 mA	4.5 V	10 20 30 40 50 60 70 80	2.5		V
	V_{OL}	3007	9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24	0.7 V	4 mA 4 mA	0.7 V	0.7 V	4 mA 4 mA	4 mA 4 mA	0.7 V	0.7 V	4 mA 4 mA			4 mA 4 mA	0.7 V	0.7 V	4 mA 4 mA	4 mA 4 mA	0.7 V	0.7 V	4 mA 4 mA		10 20 30 40 50 60 70 80 10 20 30 40 50 60 70 80		0.4	V
	V_{IC}		25 26 27 28 29 30 31 32 33 34	-18 mA		-18 mA	-18 mA			-18 mA	-18 mA			-18 mA		-18 mA	-18 mA			-18 mA	-18 mA			CLR 1D 2D 3D 4D CLK 5D 6D 7D 8D		-1.5	V
	I_{IL}	3009	35 36 37 38 39 40 41 42 43 44	0.4 V		0.4 V	0.4 V			0.4 V	0.4 V			0.4 V		0.4 V	0.4 V			0.4 V	0.4 V		5.5 V	CLR 1D 2D 3D 4D CLK 5D 6D 7D 8D	2/	2/	mA
	I_{IH1}	3010	45 46 47 48 49 50 51 52 53 54	2.7 V		2.7 V	2.7 V			2.7 V	2.7 V			2.7 V		2.7 V	2.7 V			2.7 V	2.7 V			CLR 1D 2D 3D 4D CLK 5D 6D 7D 8D		20	μA

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, or low ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases B, S, 2 Test no.	1 CLR	2 1Q	3 1D	4 2D	5 2Q	6 3Q	7 3D	8 4D	9 4Q	10 GND	11 CLK	12 5Q	13 5D	14 6D	15 6Q	16 7Q	17 7D	18 8D	19 8Q	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{IH2}	3010	55	5.5 V		5.5 V	5.5 V	"		5.5 V	5.5 V		GND	5.5 V		5.5 V	5.5 V			5.5 V	5.5 V		5.5 V	CLR		100	µA
			56																					1D			
			57																					2D			
			58																					3D			
			59																					4D			
			60																					CLK			
			61																					5Q			
			62																					6Q			
			63																					7Q			
			64																					8Q			
	I_{OS}	3011	65	4.5 V	GND	4.5 V	4.5 V	GND	GND	4.5 V	4.5 V	GND		3/	GND	4.5 V	4.5 V	GND	GND	4.5 V	4.5 V	GND		1D	-15	-100	mA
			66																					2D			
			67																					3D			
			68																					4D			
			69																					CLK			
			70																					5Q			
			71																					6Q			
			72																					7Q			
																								8Q			
	I_{CC}	3005	73	"		4.5 V	4.5 V			4.5 V	4.5 V		"	"		4.5 V	4.5 V			4.5 V	4.5 V		"	V_{CC}		27	"
2	Same tests, terminal conditions and limits as subgroup 1 except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions and limits as subgroup 1 except T_C = -55°C and V_{IC} tests are omitted.																										
7 4/ 5/	Truth table tests	3014	74	B	L	A	B	A	L	L	A	A	L	GND	A	L	A	A	L	L	A	A	L	5.0 V	6/			
			75	A	H	B	A	B	H	W	B	B	H		0	H	B	B	H	H	B	B	H				
			76		L				L	L			L		A				L				L				
			77		H				H	W			H		B				H				H				
			78		L				L	L			L		A				L				L				
			79		H				H	W			H		B				H				H				
			80		L				L	L			L		A				L				L				
			81		H				H	W			H		B				H				H				
			82		L				L	L			L		A				L				L				
			83		H				H	W			H		B				H				H				
			84		L				L	L			L		A				L				L				
			85		H				H	W			H		B				H				H				
			86		L				L	L			L		A				L				L				
			87		H				H	W			H		B				H				H				
8	Same tests and terminal conditions as subgroup 7 except T_C = +125°C and -55°C.																										
9 T_C = +25°C	f_{MAX} 7/	3003 See Fig. 4	88	4.5 V	OUT	IN	IN	OUT	OUT	IN	IN	OUT	GND	IN	OUT	IN	IN	OUT	OUT	IN	IN	OUT	5.0 V	1D	30		MHz
			89																					2D			
			90																					3D			
			91																					4D			
			92																					CLK			
			93																					5Q			
			94																					6Q			
			95																					7Q			
																								8Q			

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, or low ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A, C, 2 Test no.	1 CLR	2 1Q	3 1D	4 2D	5 2Q	6 3Q	7 3D	8 4Q	9 4Q	10 GND	11 CLK	12 5Q	13 5D	14 6D	15 6Q	16 7Q	17 7D	18 8D	19 8Q	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PHL1}	3003 Fig. 4	96	IN	OUT	4.5 V	4.5 V	OUT	OUT	4.5 V	4.5 V	OUT	GND	IN	OUT	4.5 V	4.5 V	OUT	OUT	4.5 V	4.5 V	OUT	5.0 V	CLR to 1Q	5	32	ns
			97																					CLR to 2Q			
			98																					CLR to 3Q			
			99																					CLR to 4Q			
			100																					CLR to 5Q			
			101																					CLR to 6Q			
			102																					CLR to 7Q			
			103																					CLR to 8Q			
	t_{PLH2}		104	4.5 V	OUT	IN	IN	OUT	OUT	IN	IN	OUT			OUT	IN	IN	OUT	OUT	IN	IN	OUT		CLK to 1Q			
			105																					CLK to 2Q			
			106																					CLK to 3Q			
			107																					CLK to 4Q			
			108																					CLK to 5Q			
			109																					CLK to 6Q			
			110																					CLK to 7Q			
			111																					CLK to 8Q			
	t_{PHL2}		112		OUT	IN	IN	OUT	OUT	IN	IN	OUT			OUT	IN	IN	OUT	OUT	IN	IN	OUT		CLK to 1Q			
			113																					CLK to 2Q			
			114																					CLK to 3Q			
			115																					CLK to 4Q			
			116																					CLK to 5Q			
			117																					CLK to 6Q			
			118																					CLK to 7Q			
			119																					CLK to 8Q			
																									25		MHz
10 T_C = +125°C	f_{MAX}	Same tests and terminal conditions as subgroup 9 except T_C = +125°C.																							5	42	ns
	t_{PHL1}																										
	t_{PLH2}																										
	t_{PHL2}																										
11 T_C = -55°C		Same tests, terminal conditions and limits as subgroup 10 except T_C = -55°C.																									

1/ --2.6 V minimum/5.5 V maximum.
0 V

2/ I_{IL} limits in µA are as follows:

Parameter	Circuit A	Circuit F	Circuit B	Circuit D	Circuit E
Clear	-130/-360	-105/-345	-30/-300	-120/-360	-120/-360
Clock	-120/-360	-160/-400	-30/-300	-180/-400	-120/-360
Data	-100/-340	-160/-400	-120/-360	-100/-340	-120/-360

3/ 2.5 V minimum/5.5 V maximum
0 V

4/ Input voltages shown are: A = 3.0 volts minimum and B = 0.0 V or GND.

5/ Tests shall be performed in sequence, attributes data only.

6/ Outputs: H > 1.5 V, L < 1.5 V.

7/ f_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥ 2.0 V, or low ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, Z Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				O/C	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	G	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}				
T_C = 1 +25°C	V_{OH}	3006	1	0.7 V	-1 mA	2.0 V	2.0 V	-1 mA	-1 mA	2.0 V	2.0 V	-1 mA	GND	2.0 V	-1 mA	2.0 V	2.0 V	-1 mA	-1 mA	2.0 V	2.0 V	-1 mA	4.5 V	1Q	2.4		V
			2																					2Q			
			3																					3Q			
			4																					4Q			
			5																					5Q			
			6																					6Q			
			7																					7Q			
			8																					8Q			
	V_{OL}	3007	9		12 mA	0.7 V	0.7 V	12 mA	12 mA	0.7 V	0.7 V	12 mA			12 mA	0.7 V	0.7 V	12 mA	12 mA	0.7 V	0.7 V	12 mA		1Q		0.4	
			10																					2Q			
			11																					3Q			
			12																					4Q			
			13																					5Q			
			14																					6Q			
			15																					7Q			
			16																					8Q			
	V_{IC}		17	-18 mA		-18 mA	-18 mA			-18 mA	-18 mA			-18 mA		-18 mA	-18 mA			-18 mA	-18 mA			O/C		-1.5	
			18																					1Q			
			19																					2Q			
			20																					3Q			
			21																					4Q			
			22																					5Q			
			23																					6Q			
			24																					7Q			
			25																					8Q			
			26																								
	I_{IL}	3009	27	0.4 V		0.4 V	0.4 V			0.4 V	0.4 V			0.4 V		0.4 V	0.4 V			0.4 V	0.4 V		5.5 V	O/C	1/	1/	µA
			28																					1Q			
			29																					2Q			
			30																					3Q			
			31																					4Q			
			32																					5Q			
			33																					6Q			
			34																					7Q			
			35																					8Q			
			36																								
	I_{IH1}	3010	37	2.7 V		2.7 V	2.7 V			2.7 V	2.7 V			2.7 V		2.7 V	2.7 V			2.7 V	2.7 V			O/C		20	
			38																					1Q			
			39																					2Q			
			40																					3Q			
			41																					4Q			
			42																					5Q			
			43																					6Q			
			44																					7Q			
			45																					8Q			
			46																								
	I_{IH2}		47	5.5 V		5.5 V	5.5 V			5.5 V	5.5 V			5.5 V		5.5 V								O/C		100	
			48																					1Q			
			49																					2Q			
			50																					3Q			
			51																					4Q			
			52																					5Q			
			53																					6Q			

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high $\geq$ 2.0 V, or low $\leq$ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases B, S, 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				B/C	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	6	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}				
T_C = 1 +25°C	I_{IH2}	3010	54 55 56										GND				5.5 V			5.5 V	5.5 V		5.5 V	60 70 80		100	µA
	I_{OS}	3011	57 58 59 60 61 62 63 64	GND	GND	4.5 V	4.5 V	GND	GND	4.5 V	4.5 V	GND		5.5 V	GND	4.5 V	4.5 V	GND	GND	4.5 V	4.5 V	GND		10 20 30 40 50 60 70 80	-10 2/	-120 2/	mA
	I_{CC}	3005	65	4.5 V										4.5 V										V_{CC}		40	mA
	I_{DZH}		66 67 68 69 70 71 72 73		2.7 V	2.0 V	2.0 V	2.7 V	2.7 V	2.0 V	2.0 V	2.7 V			2.7 V	2.0 V	2.0 V	2.7 V	2.7 V	2.0 V	2.0 V	2.7 V		10 20 30 40 50 60 70 80		20	µA
	I_{DZL}		74 75 76 77 78 79 80 81		0.4 V	0.7 V	0.7 V	0.4 V	0.4 V	0.7 V	0.7 V	0.4 V			0.4 V	0.7 V	0.7 V	0.4 V	0.4 V	0.7 V	0.7 V	0.4 V		10 20 30 40 50 60 70 80		-20	µA
2	Same tests, terminal conditions and limits as subgroup 1 except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions and limits as subgroup 1 except T_C = -55°C and V_{IC} tests are omitted.																										
7 3/ 4/	Truth table tests	3014	82 83 84 85 86 87 88	B	H L H H	A B B A	A B B A	H L H H	H L H H	A B B A	A B B A	H L H H	GND	A A B B	H L H H	A B B A	A B B A	H L H H	H L H H	A B B A	A B B A	H L H H	5.0 V	5/			
8	Same tests and terminal conditions as subgroup 7 except T_C = +125°C and -55°C.																										
9 T_C = +25°C	t_{PLH1}	3003 fig. 6	89 90 91 92 93 94 95 96	GND	OUT	IN	IN	OUT	OUT	IN	IN	OUT	GND	IN	OUT	IN	IN	OUT	OUT	IN	IN	OUT	5.0 V	10 to 10 20 to 20 30 to 30 40 to 40 50 to 50 60 to 60 70 to 70 80 to 80	3	18	ns

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, or low ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A, C, 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				O/C	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	G	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}				
9 T_C = +25°C	t_{PHL1}	3003 Fig. 4	97 98 99 100 101 102 103 104	GND " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	GND " " " " " " "	IN " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	5.0 V " " " " " " "	1D to 1Q 2D to 2Q 3D to 3Q 4D to 4Q 5D to 5Q 6D to 6Q 7D to 7Q 8D to 8Q	3 " " " " " " "	18 " " " " " " "	ns " " " " " " "
	t_{PLH2}	" " " " " " " "	105 106 107 108 109 110 111 112	" " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	" " " " " " " "	" " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	" " " " " " " "	G to 1Q G to 2Q G to 3Q G to 4Q G to 5Q G to 6Q G to 7Q G to 8Q	" " " " " " " "	30 " " " " " " "	" " " " " " " "
	t_{PHL2}	" " " " " " " "	113 114 115 116 117 118 119 120	" " " " " " " "	OUT " " " " " " "	GND " " " " " " "	GND " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	GND " " " " " " "	GND " " " " " " "	OUT " " " " " " "	" " " " " " " "	" " " " " " " "	OUT " " " " " " "	GND " " " " " " "	GND " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	GND " " " " " " "	GND " " " " " " "	OUT " " " " " " "	" " " " " " " "	G to 1Q G to 2Q G to 3Q G to 4Q G to 5Q G to 6Q G to 7Q G to 8Q	" " " " " " " "	" " " " " " " "	" " " " " " " "
	t_{PZH}	" " " " " " " "	121 122 123 124 125 126 127 128	IN " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	" " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	" " " " " " " "	O/C to 1Q O/C to 2Q O/C to 3Q O/C to 4Q O/C to 5Q O/C to 6Q O/C to 7Q O/C to 8Q	" " " " " " " "	28 " " " " " " "	" " " " " " " "
	t_{PZL}	" " " " " " " "	129 130 131 132 133 134 135 136	" " " " " " " "	OUT " " " " " " "	GND " " " " " " "	GND " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	GND " " " " " " "	GND " " " " " " "	OUT " " " " " " "	" " " " " " " "	" " " " " " " "	OUT " " " " " " "	GND " " " " " " "	GND " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	GND " " " " " " "	GND " " " " " " "	OUT " " " " " " "	" " " " " " " "	O/C to 1Q O/C to 2Q O/C to 3Q O/C to 4Q O/C to 5Q O/C to 6Q O/C to 7Q O/C to 8Q	" " " " " " " "	36 " " " " " " "	" " " " " " " "
	t_{PHZ}	" " " " " " " "	137 138 139 140 141 142 143 144	" " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	" " " " " " " "	" " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	" " " " " " " "	O/C to 1Q O/C to 2Q O/C to 3Q O/C to 4Q O/C to 5Q O/C to 6Q O/C to 7Q O/C to 8Q	" " " " " " " "	31 " " " " " " "	" " " " " " " "

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, or low ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, 2 Test no.	1 O/C	2 1Q	3 1D	4 2D	5 2Q	6 3Q	7 3D	8 4D	9 4Q	10 GND	11 G	12 5Q	13 5D	14 6D	15 6Q	16 7Q	17 7D	18 8D	19 8Q	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PLZ}	3003 Fig. 4	145	IB	OUT	GND	GND	OUT	OUT	GND	GND	OUT	GND	4.5 V	OUT	GND	GND	OUT	OUT	GND	GND	OUT	5.0 V	O/C to 1Q	3	30	ns
			146	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	O/C to 2Q	"	"	"
			147	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	O/C to 3Q	"	"	"
			148	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	O/C to 4Q	"	"	"
			149	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	O/C to 5Q	"	"	"
			150	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	O/C to 6Q	"	"	"
			151	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	O/C to 7Q	"	"	"
			152	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	O/C to 8Q	"	"	"
10 T_C = +125°C	t_{PLH1}	Same tests and terminal conditions as subgroup 9 except T_C = +125°C.																							"	24	"
	t_{PHL1}																								"	24	"
	t_{PLH2}																								"	39	"
	t_{PHL2}																								"	39	"
	t_{PZH}																								"	37	"
	t_{PZL}																								"	47	"
	t_{PHZ}																								"	35	"
	t_{PLZ}																								"	39	"
11 T_C = -55°C		Same tests, terminal conditions and limits as subgroup 10 except T_C = -55°C.																									

1/ I_{IL} limits as μA are as follows:

Parameter	Circuit A	B	F	C	D	E
O/C	-10/-200	-1/-100	-160/-400	-180/-420	-400 max	0/-400
G	-400 max	-1/-100	-400 max	-160/-400	-400 max	0/-400
D	-120/-360	-1/-100	-105/-345	-160/-400	-120/-360	-120/-360

2/ For circuits C and D, I_{OS} limits shall be -15/-100 mA. For circuit E, I_{OS} limits shall be -40/-225 mA.

3/ A = 3.0 V minimum. B = 0.0 V or GND.

4/ Tests shall be performed in sequence, attributes data only.

5/ Output voltages shall be:

H > 1.5 V
L < 1.5 V

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be high ≥ 2.0 V, or low ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, 2 Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				O/C	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	CLK	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}				
T_C = +25°C 1/	V_{OH}	3006	1	0.7 V	-1 mA	2.0 V	2.0 V	-1 mA	-1 mA	2.0 V	2.0 V	-1 mA	GND	1/	-1 mA	2.0 V	2.0 V	-1 mA	-1 mA	2.0 V	2.0 V	-1 mA	4.5 V	1Q	2.4		V
			2																					2Q			
			3																					3Q			
			4																					4Q			
			5																					5Q			
			6																					6Q			
			7																					7Q			
			8																					8Q			
	V_{OL}	3007	9		12 mA	0.7 V	0.7 V	12 mA	12 mA	0.7 V	0.7 V	12 mA			12 mA	0.7 V	0.7 V	12 mA	12 mA	0.7 V	0.7 V	12 mA		1Q		0.4	V
			10																					2Q			
			11																					3Q			
			12																					4Q			
			13																					5Q			
			14																					6Q			
			15																					7Q			
			16																					8Q			
	V_{IC}		17	-18 mA		-18 mA	-18 mA			-18 mA	-18 mA			-18 mA		-18 mA	-18 mA			-18 mA	-18 mA			O/C		-1.5	V
			18																					1D			
			19																					2D			
			20																					3D			
			21																					4D			
			22																					CLK			
			23																					5D			
			24																					6D			
			25																					7D			
			26																					8D			
	I_{IL}	3009	27	0.4 V		0.4 V	0.4 V			0.4 V	0.4 V			0.4 V		0.4 V	0.4 V			0.4 V	0.4 V		5.5 V	O/C	2/	2/	uA
			28																					1D			
			29																					2D			
			30																					3D			
			31																					4D			
			32																					CLK			
			33																					5D			
			34																					6D			
			35																					7D			
			36																					8D			
	I_{IH1}	3010	37	2.7 V		2.7 V	2.7 V			2.7 V	2.7 V			2.7 V		2.7 V	2.7 V			2.7 V	2.7 V			O/C		20	uA
			38																					1D			
			39																					2D			
			40																					3D			
			41																					CLK			
			42																					4D			
			43																					5D			
			44																					6D			
			45																					7D			
			46																					8D			
	I_{IH2}		47	5.5 V		5.5 V	5.5 V			5.5 V	5.5 V			5.5 V		5.5 V								O/C		100	uA
			48																					1D			
			49																					2D			
			50																					3D			
			51																					4D			
			52																					CLK			
			53																					5D			

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, or low ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, 2 Test no.	1 O/C	2 10	3 1D	4 2D	5 2Q	6 3Q	7 3D	8 4D	9 4Q	10 GND	11 CLK	12 5Q	13 5D	14 6D	15 6Q	16 7D	17 7Q	18 8D	19 8Q	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{IH2}	3010	54 55 56										GND				5.5 V			5.5 V	5.5 V		5.5 V	10 20 80		100	µA
	I_{OS}	3011	57 58 59 60 61 62 63 64	GND	GND	4.5 V	4.5 V	GND	GND	4.5 V	4.5 V	GND		1/	GND	4.5 V	4.5 V	GND	GND	4.5 V	4.5 V	GND		10 20 30 40 50 60 70 80	-30 1/	-130 1/	mA
	I_{CC}	3005	65	4.5 V																			V_{CC}		60	
	I_{OZH}		66 67 68 69 70 71 72 73		2.7 V	2.0 V	2.0 V	2.7 V	2.7 V	2.0 V	2.0 V	2.7 V			2.7 V	2.0 V	2.0 V	2.7 V	2.7 V	2.0 V	2.0 V	2.7 V		10 20 30 40 50 60 70 80		20	µA
	I_{OZL}		74 75 76 77 78 79 80 81		0.4 V	0.7 V	0.7 V	0.4 V	0.4 V	0.7 V	0.7 V	0.4 V			0.4 V	0.7 V	0.7 V	0.4 V	0.4 V	0.7 V	0.7 V	0.4 V		10 20 30 40 50 60 70 80		-20	
2	Same tests, terminal conditions and limits as subgroup 1 except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions and limits as subgroup 1 except T_C = -55°C and V_{IC} tests are omitted.																										
7 4/ 5/	Truth table tests	3014	82 83 84 85 86 87 88	a	X6/ H L L H	A A a A A	A A B A A	X6/ H L L H	X6/ H L L H	A A B A A	A A B A A	X6/ H L L H	GND	B A A B A B A	X6/ H L L H	A A B A A	A A B A A	X6/ H L L H	X6/ H L L H	A A B A A	A A B A A	X6/ H L L H	5.0 V	7/			
8	Same tests and terminal conditions as subgroup 7 except T_C = +125°C and -55°C.																										
9 T_C = +25°C	f_{MAX} 6/	3003 Fig. 4	89 90 91 92 93 94 95 96	GND	OUT	IN	IN	OUT	OUT	IN	IN	OUT	GND	IN	OUT	IN	IN	OUT	OUT	IN	IN	OUT	5.0 V	10 20 40 50 70 80	35		MHz

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, or low ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	R, S, 2 Test no.	1 O/C	2 1Q	3 1D	4 2D	5 2Q	6 3Q	7 3D	8 4D	9 4Q	10 GND	11 CLK	12 5Q	13 5D	14 6D	15 6Q	16 7Q	17 7D	18 8D	19 8Q	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PLHZ}	3003 fig. 4	97 98 99 100 101 102 103 104	GND " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	GND " " " " " " "	IN " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	5.0 V " " " " " " "	C to 1Q C to 2Q C to 3Q C to 4Q C to 5Q C to 6Q C to 7Q C to 8Q	3 " " " " " " "	28 " " " " " " "	ns " " " " " " "
	t_{PHLZ}	" " " " " " " "	105 106 107 108 109 110 111 112	" " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	" " " " " " " "	" " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	" " " " " " " "	C to 1Q C to 2Q C to 3Q C to 4Q C to 5Q C to 6Q C to 7Q C to 8Q	" " " " " " " "	" " " " " " " "	" " " " " " " "
	t_{PZH}	" " " " " " " "	113 114 115 116 117 118 119 120	IN " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	" " " " " " " "	1/ " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	" " " " " " " "	O/C to 1Q O/C to 2Q O/C to 3Q O/C to 4Q O/C to 5Q O/C to 6Q O/C to 7Q O/C to 8Q	" " " " " " " "	" " " " " " " "	" " " " " " " "
	t_{PZL}	" " " " " " " "	121 122 123 124 125 126 127 128	" " " " " " " "	OUT " " " " " " "	GND " " " " " " "	GND " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	GND " " " " " " "	GND " " " " " " "	OUT " " " " " " "	" " " " " " " "	" " " " " " " "	OUT " " " " " " "	GND " " " " " " "	GND " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	GND " " " " " " "	GND " " " " " " "	OUT " " " " " " "	" " " " " " " "	O/C to 1Q O/C to 2Q O/C to 3Q O/C to 4Q O/C to 5Q O/C to 6Q O/C to 7Q O/C to 8Q	" " " " " " " "	" " " " " " " "	" " " " " " " "
	t_{PHZ}	" " " " " " " "	129 130 131 132 133 134 135 136	" " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	" " " " " " " "	" " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	" " " " " " " "	O/C to 1Q O/C to 2Q O/C to 3Q O/C to 4Q O/C to 5Q O/C to 6Q O/C to 7Q O/C to 8Q	" " " " " " " "	32 " " " " " " "	" " " " " " " "
	t_{PLZ}	" " " " " " " "	137 138 139 140 141 142 143 144	" " " " " " " "	OUT " " " " " " "	GND " " " " " " "	GND " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	GND " " " " " " "	GND " " " " " " "	OUT " " " " " " "	" " " " " " " "	" " " " " " " "	OUT " " " " " " "	GND " " " " " " "	GND " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	GND " " " " " " "	GND " " " " " " "	OUT " " " " " " "	" " " " " " " "	O/C to 1Q O/C to 2Q O/C to 3Q O/C to 4Q O/C to 5Q O/C to 6Q O/C to 7Q O/C to 8Q	" " " " " " " "	30 " " " " " " "	" " " " " " " "

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, or low ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A, C, E Test no.	1 O/C	2 1Q	3 1D	4 2D	5 2Q	6 3Q	7 3D	8 4D	9 4Q	10 GND	11 CLR	12 5Q	13 5D	14 6D	15 6Q	16 7Q	17 7D	18 8D	19 8Q	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
10 T_C = +125°C	f_{MAX}	Same tests and terminal conditions as subgroup 9 except T_C = +125°C.																						26		MHz
	t_{PLH2}																							3	37	ns
	t_{PHL2}																							-	-	-
	t_{PZH}																							-	-	-
	t_{PZL}																							-	-	-
	t_{PHZ}																							-	-	-
	t_{PLZ}																							-	39	-
11 T_C = -55°C		Same tests, terminal conditions and limits as subgroup 10 except T_C = -55°C.																								

1/ Apply ⊓ 2.5 V min/5.5 V max. Pulse prior to test.
0 V

2/ I_{IL} limits in µA are as follows:

Parameter	Circuit A	F	B	C	D	E
O/C	-10/-200	-160/-400	-400 max	-180/-420	-400 max	0/-400
Clock	-400 max	-400 max	-400 max	-160/-400	-400 max	0/-400
Data	-75/-250	-75/-250	-400 max	-100/-340	-75/-250	-120/-360

3/ For circuits C and D, I_{OS} limits shall be -15/-100 mA. For circuit E, I_{OS} limits shall be -40/-225 mA.

4/ A = 3.0 V min. B = 0.0 V or GND.

5/ Tests shall be performed in sequence, attributes data only.

6/ X = Indeterminate output voltage.

7/ Output voltages shall be either:

H > 1.5 V
L < 1.5 V

8/ f_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated may be high ≥ 2.0 V, or low ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				G	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	CLK	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1 2 3 4 5 6 7 8	0.7 V	-.4 mA	2.0 V	2.0 V	-.4 mA	-.4 mA	2.0 V	2.0 V	-.4 mA	GND	1/	-.4 mA	2.0 V	2.0 V	-.4 mA	-.4 mA	2.0 V	2.0 V	-.4 mA	4.5 V	1Q 2Q 3Q 4Q 5Q 6Q 7Q 8Q	2.5		V
	V_{OL}	3007	9 10 11 12 13 14 15 16		4 mA	0.7 V	0.7 V	4 mA	4 mA	0.7 V	0.7 V	4 mA			4 mA	0.7 V	0.7 V	4 mA	4 mA	0.7 V	0.7 V	4 mA		1Q 2Q 3Q 4Q 5Q 6Q 7Q 8Q		0.4	
	V_{IC}		17 18 19 20 21 22 23 24 25 26	-18 mA		-18 mA	-18 mA			-18 mA	-18 mA			-18 mA		-18 mA	-18 mA			-18 mA	-18 mA			1D 2D 3D 4D CLK 5D 6D 7D 8D		-1.5	
	I_{IL}	3009	27 28 29 30 31 32 33 34 35 36	0.4 V		0.4 V	0.4 V			0.4 V	0.4 V			0.4 V		0.4 V	0.4 V			0.4 V	0.4 V		5.5 V	1D 2D 3D 4D CLK 5D 6D 7D 8D	2/	2/	-A
	I_{IH1}	3010	37 38 39 40 41 42 43 44 45 46	2.7 V		2.7 V	2.7 V			2.7 V	2.7 V			2.7 V		2.7 V	2.7 V			2.7 V	2.7 V			1D 2D 3D 4D CLK 5D 6D 7D 8D		20	
	I_{IH2}		47 48 49 50 51 52 53	5.5 V		5.5 V	5.5 V			5.5 V	5.5 V			5.5 V		5.5 V								1D 2D 3D 4D CLK 5D 6D 7D 8D		100	

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, or low ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case R, S, 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				G	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	CLK	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}				
1 T_C = +25°C	I_{IH2}	3010	54										GND				5.5 V			5.5 V	5.5 V		5.5 V	6Q 7Q 8Q		100	µA
			55										"										"				
			56										"										"				
	I_{OS}	3011	57	0.7 V	GND	4.5 V	4.5 V	GND	GND	4.5 V	4.5 V	GND	"	3/	GND	4.5 V	4.5 V	GND	GND	4.5 V	4.5 V	GND	"	1Q	-15	-100	mA
			58	"									"	"									"	2Q			
			59	"									"	"									"	3Q			
			60	"									"	"									"	4Q			
			61	"									"	"									"	5Q			
			62	"									"	"									"	6Q			
			63	"									"	"									"	7Q			
			64	"									"	"									"	8Q			
	I_{CC}	3005	65	GND		GND	GND			GND	GND		"	"		GND	GND			GND	GND		"	V_{CC}		32	"
2	Same tests, terminal conditions and limits as subgroup 1 except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions and limits as subgroup 1 except T_C = -55°C and V_{IC} tests are omitted.																										
7 4/ 5/	Truth table tests	3014	66	B	x 6/	B	B	x 6/	x 6/	B	B	x 6/	GND	B	x 6/	B	B	x 6/	x 6/	B	B	x 6/	5.0 V	7/			
			67	A	L	A	A	L	L	A	A	H	"	A	H	A	A	H	L	A	A	L	"				
			68	B	H	B	B	H	H	B	B	L	"	B	L	B	B	L	H	B	B	H	"				
			69	A	L	A	A	L	L	A	A	H	"	A	H	A	A	H	L	A	A	L	"				
			70	B	H	B	B	H	H	B	B	L	"	B	L	B	B	L	H	B	B	H	"				
			71	A	L	A	A	L	L	A	A	H	"	A	H	A	A	H	L	A	A	L	"				
			72	B	H	B	B	H	H	B	B	L	"	B	L	B	B	L	H	B	B	H	"				
			73	A	L	A	A	L	L	A	A	H	"	A	H	A	A	H	L	A	A	L	"				
			74	B	H	B	B	H	H	B	B	L	"	B	L	B	B	L	H	B	B	H	"				
			75	A	L	A	A	L	L	A	A	H	"	A	H	A	A	H	L	A	A	L	"				
			76	B	H	B	B	H	H	B	B	L	"	B	L	B	B	L	H	B	B	H	"				
			77	A	L	A	A	L	L	A	A	H	"	A	H	A	A	H	L	A	A	L	"				
			78	B	H	B	B	H	H	B	B	L	"	B	L	B	B	L	H	B	B	H	"				
			79	A	L	A	A	L	L	A	A	H	"	A	H	A	A	H	L	A	A	L	"				
			80	B	H	B	B	H	H	B	B	L	"	B	L	B	B	L	H	B	B	H	"				
			81	A	L	A	A	L	L	A	A	H	"	A	H	A	A	H	L	A	A	L	"				
			82	B	H	B	B	H	H	B	B	L	"	B	L	B	B	L	H	B	B	H	"				
8	Same tests and terminal conditions as subgroup 7 except T_C = +125°C and -55°C.																										
9 T_C = +25°C	f_{MAX} 8/	3003 See Fig. 4	83	GND	OUT	IN	IN	OUT	OUT	IN	IN	OUT	GND	IN	OUT	IN	IN	OUT	OUT	IN	IN	OUT	5.0 V	CLK to Q	30		MHz
			84	"									"	"									"				
			85	"									"	"									"				
			86	"									"	"									"				
			87	"									"	"									"				
			88	"									"	"									"				
			89	"									"	"									"				
			90	"									"	"									"				
	t_{PLH2}	"	91	"	OUT	IN	IN	OUT	OUT	IN	IN	OUT	"	"	OUT	IN	IN	OUT	OUT	IN	IN	OUT	"	CLK to 1Q	5	32	ns
			92	"									"	"									"	CLK to 2Q			
			93	"									"	"									"	CLK to 3Q			
			94	"									"	"									"	CLK to 4Q			
			95	"									"	"									"	CLK to 5Q			
			96	"									"	"									"	CLK to 6Q			
			97	"									"	"									"	CLK to 7Q			
			98	"									"	"									"	CLK to 8Q			

See footnotes at end of device type 04.

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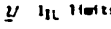
TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, or low ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A, S, 2 test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				G	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	CLK	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}				
9 T_C = +25°C	t_{PHL2}	3003 See Fig. 4	99 100 101 102 103 104 105 106	GND " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	GND " " " " " " "	IN " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	5.0 V " " " " " " "	CLK to 1Q CLK to 2Q CLK to 3Q CLK to 4Q CLK to 5Q CLK to 6Q CLK to 7Q CLK to 8Q	5 " " " " " " "	32 " " " " " " "	ns " " " " " " "
10 T_C = +125°C	f_{MAX}	Same tests and terminal conditions as subgroup 9 except T_C = 125°C.																							25		MHz
	t_{PLH2}																								5	42	ns
	t_{PHL2}																								5	42	ns
11 T_C = -55°C		Same tests, terminal conditions and limits as subgroup 10 except T_C = -55°C.																									

1/  --2.5 V minimum/5.5 V maximum.
0 V

2/ I_{IL} limits in uA are as follows:

Parameter	Circuit A	Circuit F	Circuit B	Circuit D
Enable	-120/-360	-105/-345	-30/-300	-120/-360
Clock	-120/-360	-160/-400	-30/-300	-160/-400
Data	-100/-340	-160/-400	-120/-360	-100/-340

3/  2.5 V minimum/5.5 V maximum
0 V

4/ A = 3.0 V min. B = 0.0 V or GND.

5/ Tests shall be performed in sequence, attributes data only.

6/ X = Indeterminate output voltage.

7/ Output shall be: H > 1.5 V, L < 1.5 V.

8/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design application and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirements for JAN marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential
V_{IN}	Voltage level at an input terminal
f_{MAX}	Maximum clock frequency

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuits types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54LS273
02	54LS373
03	54LS374
04	54LS377

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design application and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

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6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential
V_{IN}	Voltage level at an input terminal
f_{MAX}	Maximum clock frequency

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuits types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54LS273
02	54LS373
03	54LS374
04	54LS377

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6.6 Manufacturers' designation. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designation.

Device type	Circuits					
	A	B	C	D	F	E
	Texas Instrument Incorporated	Signetics Corporation	National Semiconductor Corporation	Raytheon Company	Motorola Incorporated	Fairchild
01	X	X		X	X	X
02	X		X	X	X	
03	X	X	X	X		
04	X	X		X	X	

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1001)

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/325C	Microcircuits, Digital, Bipolar Low-Power Schottky TTL, Flip-Flops, Cascadable, Monolithic Silicon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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